

6(a)(i)	$E = \Delta V / \Delta d$	C1
	$E = (180 + 120) / (2.0 \times 10^{-2})$ $= 1.5 \times 10^4 \text{ N C}^{-1}$	A1
6(a)(ii)	vertically downwards	B1
6(b)(i)	number of protons = 92	A1
	number of neutrons = 146	A1
	number of electrons = 90	A1
6(b)(ii)	$F = EQ$	C1
	$= 1.5 \times 10^4 \times 2 \times 1.60 \times 10^{-19}$	A1
	$= 4.8 \times 10^{-15} \text{ N}$	
6(b)(iii)	number of α -particles = 2	A1
	number of β^- particles = 2	A1